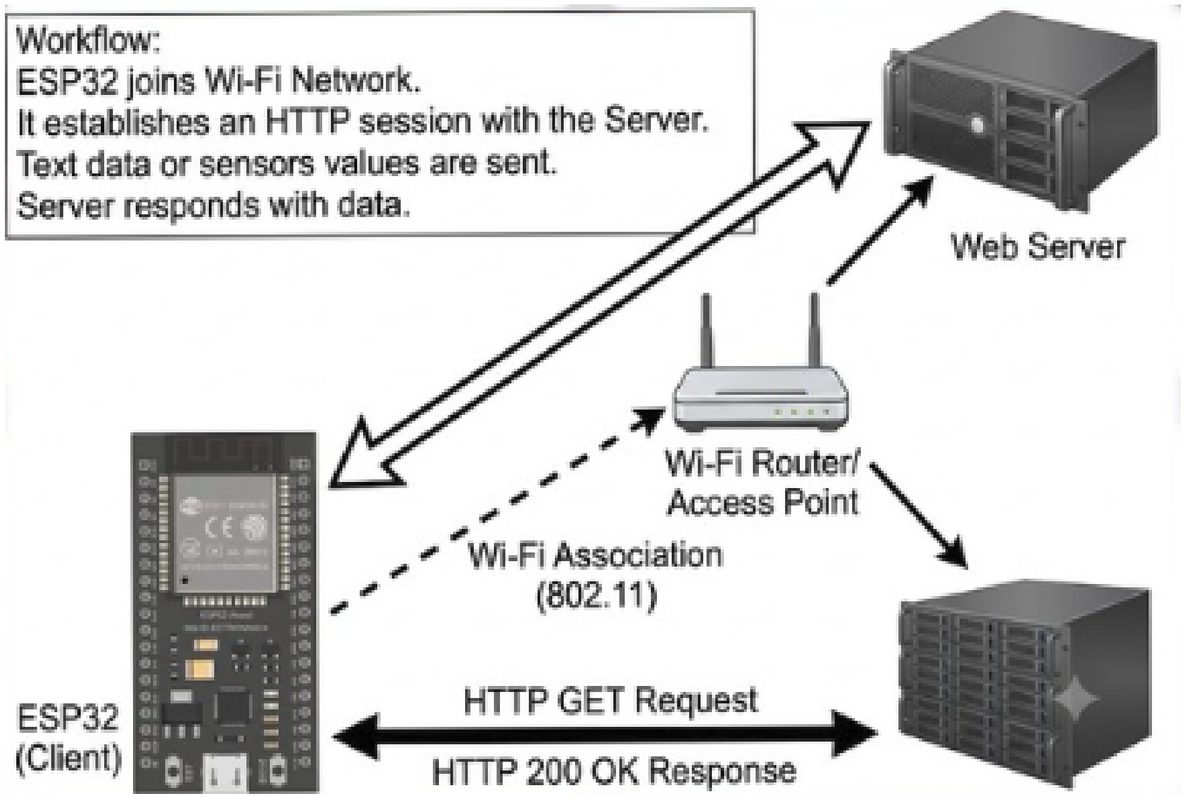
HTTP Communication Protocol

Client-Server Web Communication via Wi-Fi Network

What Is HTTP?

A request-response protocol enabling an ESP32 client to fetch data from or send data to web servers using standard Wi-Fi infrastructure.



Requires Access Point

Connects to standard 2.4 GHz Wi-Fi routers.

Standard Protocols

Operates over standard TCP/IP using <WiFi.h> and <HTTPClient.h> libraries.

Structured Data

ransmits and receives data using formatted standard structures like JSON payloads.

Client-Server Architecture

Uses distinct GET (retrieve data) and POST (transmit data) requests.

Where Is HTTP Used in IoT?

Replace peer-to-peer/mesh applications with the following

Cloud IoT Dashboards & Logging

Weather & REST API Data Retrieval

Remote Sensor Telemetry to Central Servers

Web Service Notifications & Webhooks

Prerequisites & Hardware :

Hardware: One ESP32 Development Board (e.g., DOIT ESP32 DEVKIT V1)

USB data cable: Required for programming and Serial Monitor access. Make sure the cable supports data, not only charging.

Power: USB power is sufficient for normal testing. A separate regulated supply can be used if the board's power-input requirements are satisfied.

Software: Arduino IDE installed on your computer

Network Requirements: A 2.4 GHz Wi-Fi Network (SSID and Password)

Step 1: Configure the Arduino IDE

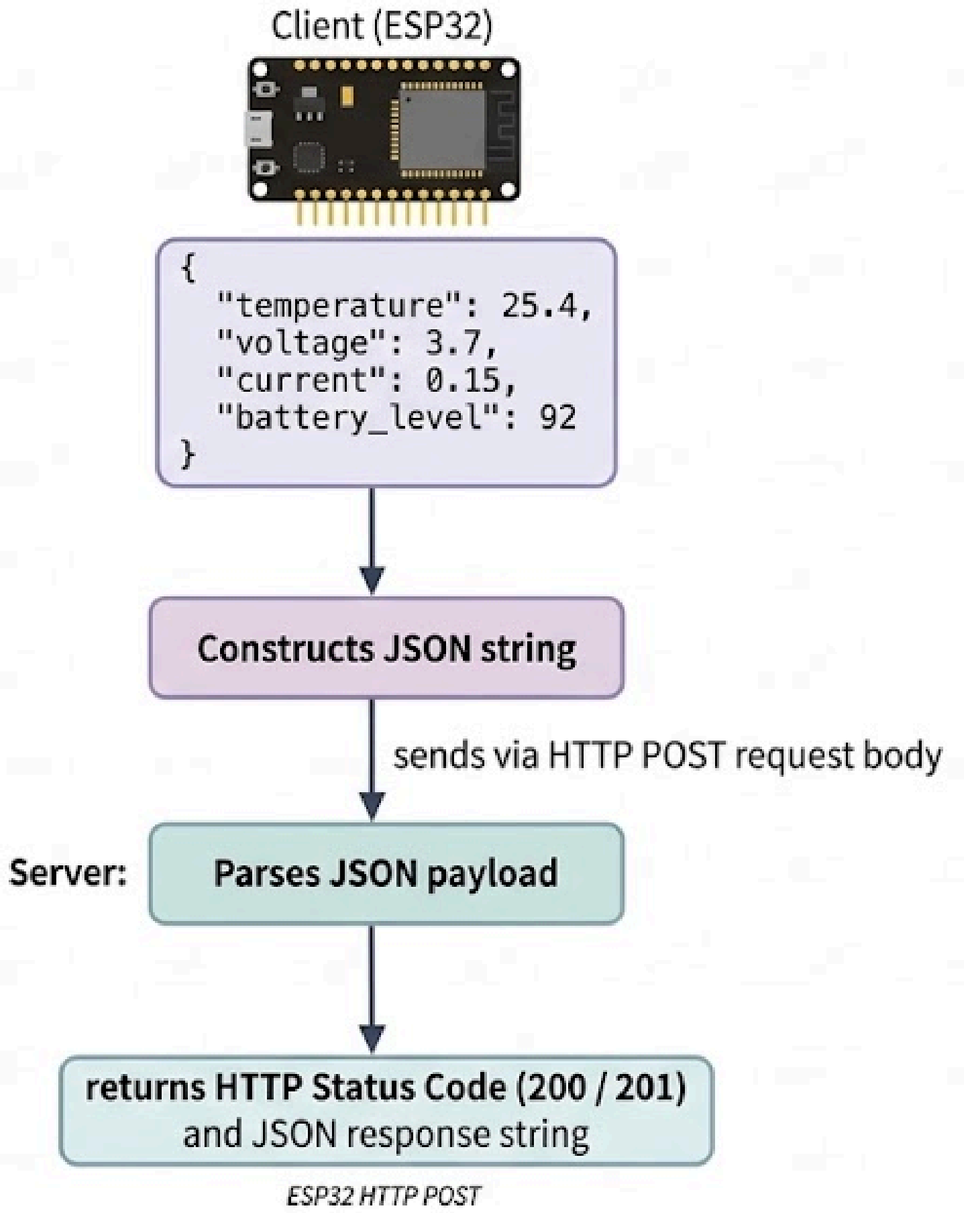
Step 2: Connect ESP32 to Wi-Fi Network

Step 3: Perform an HTTP GET Request (Download Data)

Step 4: Perform an HTTP POST Request (Send Data/Telemetry)

Step 5: Verify HTTP Response Codes (e.g., 200 OK / 201 Created)

Understand the JSON Payload Flow (HTTP POST)

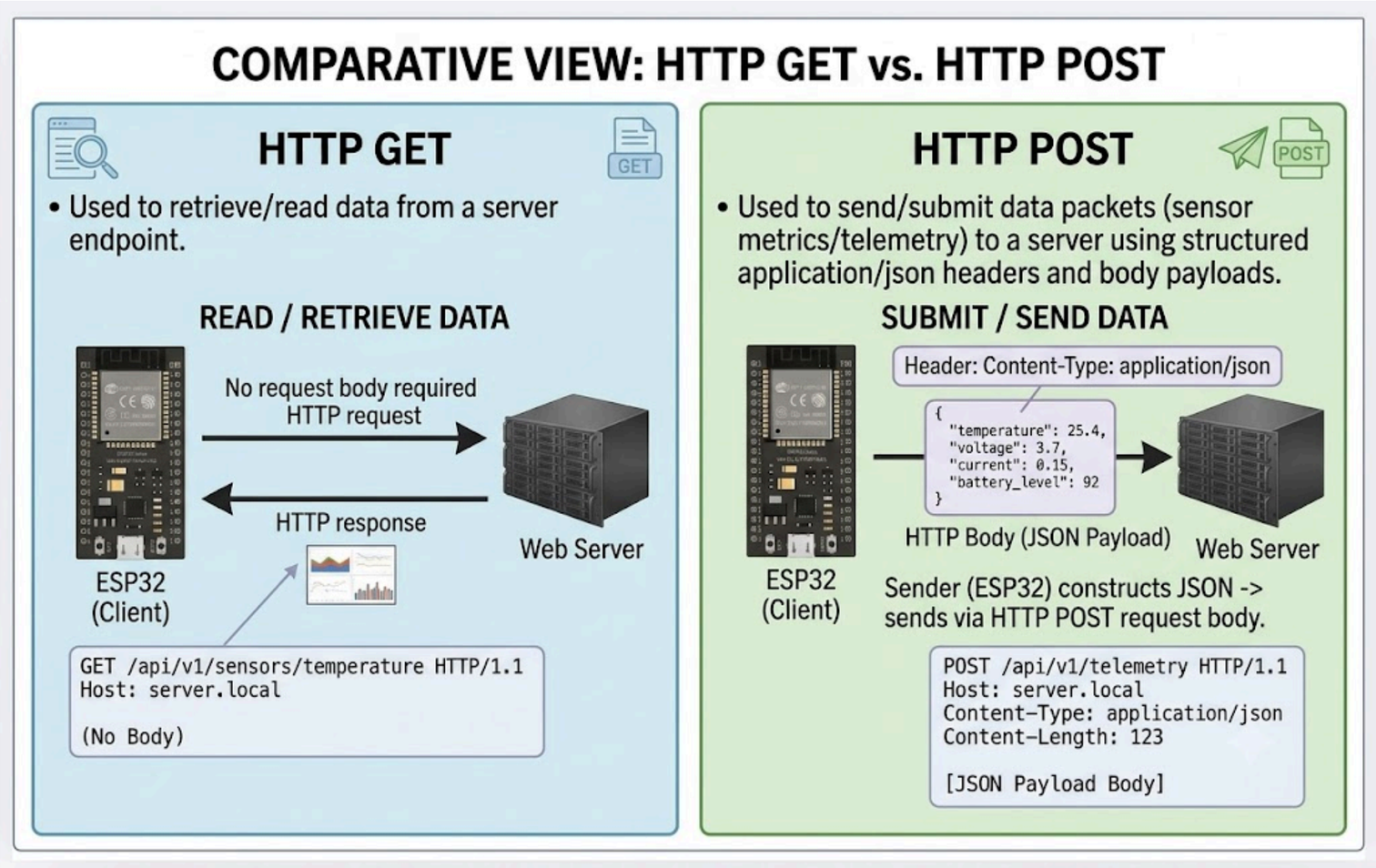


ESP32 HTTP POST

ESP32 HTTP POST chn payload flow

HTTP Methods Comparison

HTTP POST sends a request payload to a specific server endpoint URL (such as a REST API or database server) to process or store data, rather than retrieving data from an open web resource like HTTP GET.



An HTTP URL target determines the interaction type. http:// endpoints communicate in plaintext, while https:// endpoints enforce SSL/TLS encryption. Do not name a variable httpsServer and assign an unencrypted http:// URL to it; the scheme prefix determines the actual security protocol.

That's HTTP Communication!

The access the implementation guide
Scan the QR / click on the link in the description

Thank You